

# ECO387C - Fall 2026

## Discussion 6

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### 1 Introduction

Consider the stochastic neoclassical growth model, where the planner solves:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t u(c_t) \right\}$$

subject to

$$\begin{aligned} c_t + k_{t+1} &\leq z_t f(k_t) + (1 - \delta)k_t \\ \log(z_{t+1}) &= \rho \log(z_t) + \varepsilon_{t+1}, \quad \varepsilon \sim \text{I.I.D. } \mathcal{N}(0, \sigma_\varepsilon^2) \end{aligned}$$

for a given  $k_0$  and  $z_0$ . The corresponding Bellman equation is given by:

$$v(k, z) = \max_{\{0 \leq k' \leq z f(k) + (1 - \delta)k\}} \{u(z f(k) + (1 - \delta)k - k') + \beta \mathbb{E}_{z'|z}[v(k', z')]\}$$

for all  $(k, z)$ , where  $z'$  evolves according to the AR(1) process as above.

Solving this problem requires us to compute expectations, or  $\mathbb{E}_{z'|z}[v(k', z')]$ . One way is to approximate the AR(1) process, which is continuous, with a **finite-state Markov chain**.

Denote  $\pi(z' | z)$  the conditional density of  $z'$  given  $z$ . This can be approximated as

$$\begin{aligned} \mathbb{E}_{z'|z=z_j}[v(k', z')] &= \int v(k', z') \pi(z' | z) dz' \\ &\approx \sum_{\ell=1}^{N_z} v(k', z' = z_\ell) \underbrace{P(z' = z_\ell | z = z_j)}_{p_{j\ell}} \end{aligned}$$

The goal is to

1. Obtain the set of gridpoints  $Z = \{z_1, \dots, z_{N_z}\}$  and the transition probabilities  $p_{j\ell}$ , for all  $j, \ell \in \{1, \dots, N_z\}$ . In practice we discretize  $\log z$  and convert back to the level of  $z$ .
2. Given the gridpoints and transition probabilities, solve for  $v(k, z)$  using the VFI.

## 2 Discretizing AR(1) processes

### 2.1 Tauchen (1986)

Tauchen's method allows us to find a probability matrix that governs the transition on a given discrete grid that approximates an AR(1) process.<sup>1</sup> Suppose that  $\eta' \sim N(0, 1)$ , then I can write an AR(1) process as

$$z' = (1 - \rho)\bar{z} + \rho z + \sigma\eta'$$

$$\eta' = \frac{z' - \rho z - (1 - \rho)\bar{z}}{\sigma}$$

Tauchen's method is based on the fact that, conditional on  $z$ ,  $z'$  is normally distributed with mean  $(1 - \rho)\bar{z} + \rho z$  and standard deviation  $\sigma$ .

**Step 1. Grid Choice** The grid must be chosen first. Let  $\sigma_z = \sigma/\sqrt{1 - \rho^2}$  be the unconditional standard deviation of the process, pick a width parameter  $m$ , and place  $N_z$  equally spaced points on  $[\bar{z} - m\sigma_z, \bar{z} + m\sigma_z]$ :

$$z_1 = \bar{z} - m\sigma_z, \quad z_{N_z} = \bar{z} + m\sigma_z, \quad w = \frac{z_{N_z} - z_1}{N_z - 1} = \frac{2m\sigma_z}{N_z - 1},$$

$w$  is the distance between two neighbouring gridpoints.

**Step 2. Probability Assignment** Now let  $p_{j\ell} = P(z' = z_\ell | z = z_j)$  be the probability of moving from gridpoint  $z_j$  to gridpoint  $z_\ell$ , and let  $\Phi$  denote the cdf of the standard normal distribution. Tauchen assigns to  $z_\ell$  all the conditional probability mass that falls within  $\pm w/2$  of it, so for  $\ell \in \{2, \dots, N_z - 1\}$ ,

$$\begin{aligned} p_{j\ell} &= P\left(z_\ell - \frac{w}{2} \leq (1 - \rho)\bar{z} + \rho z_j + \sigma\eta' \leq z_\ell + \frac{w}{2}\right) \\ &= P\left(\frac{z_\ell - \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma} \leq \eta' \leq \frac{z_\ell + \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) \\ &= \Phi\left(\frac{z_\ell + \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) - \Phi\left(\frac{z_\ell - \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) \end{aligned}$$

For  $\ell = 1$

$$\begin{aligned} p_{j1} &= P\left((1 - \rho)\bar{z} + \rho z_j + \sigma\eta' \leq z_1 + \frac{w}{2}\right) \\ &= P\left(\eta' \leq \frac{z_1 + \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) \\ &= \Phi\left(\frac{z_1 + \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) \end{aligned}$$

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<sup>1</sup>Tauchen, George. "Finite state markov-chain approximations to univariate and vector autoregressions." *Economics letters* 20.2 (1986): 177-181.

For  $\ell = N_z$

$$\begin{aligned}
p_{jN_z} &= P\left((1-\rho)\bar{z} + \rho z_j + \sigma\eta' \geq z_{N_z} - \frac{w}{2}\right) \\
&= 1 - P\left(\eta' \leq \frac{z_{N_z} - \frac{w}{2} - (1-\rho)\bar{z} - \rho z_j}{\sigma}\right) \\
&= 1 - \Phi\left(\frac{z_{N_z} - \frac{w}{2} - (1-\rho)\bar{z} - \rho z_j}{\sigma}\right)
\end{aligned}$$

**Summary** For all  $j \in \{1, \dots, N_z\}$ , the transition probabilities  $p_{j\ell}$  are given by:

$$p_{j\ell} = \begin{cases} \Phi\left(\frac{z_\ell + \frac{w}{2} - (1-\rho)\bar{z} - \rho z_j}{\sigma}\right) & \text{if } \ell = 1, \\ \Phi\left(\frac{z_\ell + \frac{w}{2} - (1-\rho)\bar{z} - \rho z_j}{\sigma}\right) - \Phi\left(\frac{z_{\ell-1} - \frac{w}{2} - (1-\rho)\bar{z} - \rho z_j}{\sigma}\right) & \text{if } \ell \in \{2, \dots, N_z - 1\}, \\ 1 - \Phi\left(\frac{z_{N_z} - \frac{w}{2} - (1-\rho)\bar{z} - \rho z_j}{\sigma}\right) & \text{if } \ell = N_z. \end{cases}$$

## 2.2 Rouwenhorst (1995)

Consider the same AR(1) process:

$$z' = (1-\rho)\bar{z} + \rho z + \sigma\eta'$$

where  $|\rho| < 1$ , and  $\eta$  is an i.i.d. shock that follows a standard normal distribution. This AR(1) process has a stationary normal distribution with mean  $\bar{z}$  and variance  $\sigma_z^2 = \frac{\sigma^2}{1-\rho^2}$ . Rouwenhorst (1995) proposes a discretization method that matches the three key moments of the AR(1) process: the stationary mean, variance, and autocorrelation.<sup>2</sup> The method follows the following three steps.

**Step 1. Grid Choice** Define  $v = \sqrt{\frac{n-1}{1-\rho^2}}\sigma$  and create  $n$  discrete grid points over the interval  $[\bar{z}-v, \bar{z}+v]$ .  $n$  is going to determine the number of states of the Markov process.

**Step 2. Probability Assignment** Construct the Rouwenhorst probability transition matrix recursively. For  $n = 2$ , the transition matrix is:

$$P_2 = \begin{bmatrix} p & 1-p \\ 1-q & q \end{bmatrix}$$

For  $n > 2$ , the  $P_n$  matrix can be generated recursively from  $P_{n-1}$  as follows:

$$P_n = p \begin{bmatrix} P_{n-1} & \mathbf{0} \\ \mathbf{0}' & 0 \end{bmatrix} + (1-p) \begin{bmatrix} \mathbf{0} & P_{n-1} \\ 0 & \mathbf{0}' \end{bmatrix} + (1-q) \begin{bmatrix} \mathbf{0}' & 0 \\ P_{n-1} & \mathbf{0} \end{bmatrix} + q \begin{bmatrix} 0 & \mathbf{0}' \\ \mathbf{0} & P_{n-1} \end{bmatrix}$$

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<sup>2</sup>Rouwenhorst, K. Geert. "Asset pricing implications of equilibrium business cycle models." *Frontiers of Business Cycle Research*, 1 (1995): 294-330.

where  $\mathbf{0}$  is an  $(n-1) \times 1$  column vector of zeros and  $\mathbf{0}'$  is its transpose. Divide all but the top and bottom rows by 2 so that the sum of the elements of each row equals 1. For example,  $P_3$  is:

$$P_3 = \begin{bmatrix} p^2 & 2p(1-p) & (1-p)^2 \\ p(1-q) & pq + (1-p)(1-q) & q(1-p) \\ (1-q)^2 & 2q(1-q) & q^2 \end{bmatrix}$$

This construction method is particularly useful for highly persistent processes ( $\rho > 0.9$ ).<sup>3</sup> It also matches the key moments of the AR(1) process *regardless of the number of grid-points*. For example, it can be shown that  $p+q-1 = \rho$ . In most applications, it is assumed that  $p = q$ , and thus, we can simply set  $p = \frac{\rho+1}{2}$  to construct the  $P_n$ .

Furthermore, it can be shown that a Rouwenhorst chain on an evenly spaced grid of half-width  $v$  has stationary variance  $\frac{v^2}{n-1}$ , whatever  $n$  is. Matching this to the stationary variance of the AR(1) process,  $\frac{\sigma^2}{1-\rho^2}$ , and solving for  $v$  is exactly the choice  $v = \sqrt{\frac{n-1}{1-\rho^2}}\sigma$  made in the first step.

### 3 Value function iteration

We now apply the discretization to the stochastic neoclassical growth model. The Bellman equation is

$$v(k, z) = \max_{\{0 \leq k' \leq zf(k) + (1-\delta)k\}} \{u(zf(k) + (1-\delta)k - k') + \beta \mathbb{E}_{z'|z} v(k', z')\}$$

for all  $(k, z)$ . The  $\log z'$  evolves according to the AR(1) process:

$$\log(z') = \rho \log(z) + \varepsilon, \quad \varepsilon \sim \text{I.I.D. } \mathcal{N}(0, \sigma_\varepsilon^2)$$

#### 3.1 Discretizing the shock

Denote  $\tilde{z} \equiv \log(z)$ , so that the shock process becomes

$$\tilde{z}' = \rho \tilde{z} + \varepsilon, \quad \varepsilon \sim \text{I.I.D. } \mathcal{N}(0, \sigma_\varepsilon^2).$$

Either approximation method discussed in the previous section returns

- a grid  $\tilde{z} \in \{\tilde{z}_1, \dots, \tilde{z}_{N_z}\}$ , in logs, and
- a transition matrix  $P$  whose typical element is

$$p_{j\ell} = P(\tilde{z}' = \tilde{z}_\ell \mid \tilde{z} = \tilde{z}_j), \quad \sum_{\ell=1}^{N_z} p_{j\ell} = 1 \quad \text{for every } j.$$

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<sup>3</sup>Kopecky, Karen A., and Richard MH Suen. "Finite state Markov-chain approximations to highly persistent processes." *Review of Economic Dynamics*, 13.3 (2010): 701-714.

The production function depends on the *level* of productivity, so the last step is to undo the log:

$$z_j = \exp(\tilde{z}_j), \quad j = 1, \dots, N_z.$$

The transition probability matrix is unaffected by this transformation.

### 3.2 The discretized Bellman equation

Denote the set of capital grid in ascending order,  $\mathcal{K} \equiv \{k_1, \dots, k_{N_k}\}$ . The Bellman equation is as follows: for every  $k \in \mathcal{K}$  and every  $j = 1, \dots, N_z$ ,

$$v(k, z = z_j) = \max_{k' \in \mathcal{K}} \left\{ u \left( \underbrace{z_j f(k) + (1 - \delta)k - k'}_{\equiv c} \right) + \beta \sum_{\ell=1}^{N_z} p_{j\ell} v(k', z' = z_\ell) \right\}$$

Note that the  $v$  is now just an  $N_k \times N_z$  matrix. Once  $v$  has converged, consumption can be backed out from the budget constraint,

$$c(k, z = z_j) = z_j f(k) + (1 - \delta)k - k'(k, z = z_j).$$

### 3.3 Results

Check the attached code for calibration of parameters. Setting  $z = 1$  and  $k' = k$  in the Euler equation  $1 = \beta [\alpha z k^{\alpha-1} + 1 - \delta]$  gives the deterministic steady state value of capital:

$$k^* = \left( \frac{\alpha}{1/\beta - 1 + \delta} \right)^{\frac{1}{1-\alpha}},$$

Figure 1 reports the result.

- **State dependence.** A higher  $z$  shifts both policies up. Under higher productivity, the agent consumes *and* saves more.
- **Gridpoints.** A higher number of gridpoints makes the policy functions more smooth.

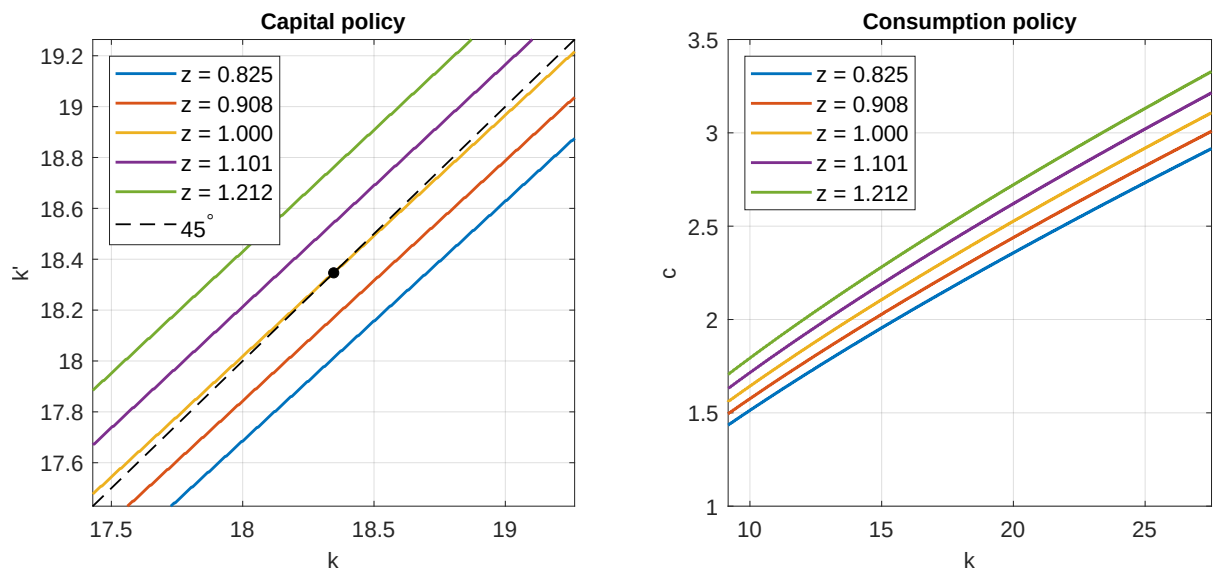


Figure 1: Policy functions from value function iteration